AQ2.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1:

1 point

Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$. If

- $P = -1M_{11} - 2M_{12} + 0M_{13}$
- $Q = -2M_{21} + 0M_{22} + 1M_{23}$
- $R = -1M_{31} - 0M_{32} + 1M_{33}$

where M_{ij} is the minor with respect to the (i, j)-th entry, then choose the set of correct options.

- [ ] $det(A) = -2$.
- [ ] $P \neq Q$.
- [ ] $P \neq Q \neq R$.
- [ ] $P = Q = R$.
- [ ] $Q \neq R$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = -2$.
$P = Q = R$.

1 point
Suppose for a real 3×3 matrix A, there exists a real 3×3 matrix P such that $D = PAP^{-1}$ is a real 3×3 diagonal matrix. Choose the correct set of options.

- [ ] $det\,(A)$ must be equal to $det\,(P)$.
- [ ] $det\,(A)$ must be equal to $det\,(D)$.
- [ ] The matrix A must be equal to D.
- [ ] If D is the identity matrix of order 3, then A must be the identity matrix of order 3

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det\,(A)$ must be equal to $det\,(D)$.
If D is the identity matrix of order 3, then A must be the identity matrix of order 3

1 point
Choose the set of correct options for square matrices of order 3.

- [ ] If $AB = 0$, then one of the matrices between A and B must have determinant 0.
- [ ] If $AB = 0$, then both the matrices A and B must have determinant 0.
- [ ] If $AB = 3I$, where I denotes the identity matrix of order 3, then $det A = \frac{3}{det(B)}$.
- [ ] If $AB = 3I$, where I denotes the identity matrix of order 3, then $det A = \frac{9}{det(B)}$.
- [ ] If $AB = 3I$, where I denotes the identity matrix of order 3, then $det A = \frac{27}{det(B)}$.
- [ ] If $AB = 3I$, where I denotes the identity matrix of order 3, then both A and B must have non-zero determinant.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If $AB = 0$, then one of the matrices between A and B must have determinant 0.
If $AB = 3I$, where I denotes the identity matrix of order 3, then $det A = \frac{27}{det(B)}$.

If $AB = 3I$, where I denotes the identity matrix of order 3, then both A and B must have non-zero determinant.

1 point

Consider a matrix $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$. If $a + b + c$ is divisible by 6 then choose the set of correct options.

- [ ] $det(A)$ is divisible by 5.
- [ ] $det(A)$ is divisible by 6.
- [ ] $det(A)$ is divisible by 2.
- [ ] $det(A)$ is divisible by 3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$det(A)$ is divisible by 6.
$det(A)$ is divisible by 2.
$det(A)$ is divisible by 3.

Let $A = [a_{ij}]$ be a square matrix of order 3, where $a_{ij} = i$. Find $det(A)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 0
1 point

Level 2:

Let $A = [\delta_{ij}]$A = [δij] be a square matrix of order 33, where $\delta_{ij} = i \times j$δij = i × j. Find $det(A)$ det(A).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
1 point
Consider the following square matrices:

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

A = [201320162019 201420172020201520222021] , B = [201620132019201720142020202220152021] and C = [403220132019403420142020404420152021]

Choose the set of correct options.

- [ ] $det(B) = det(A)$det(B) = det(A).
- [ ] $det(A) = -det(B)$det(A) = −det(B).
- [ ] $det(C) \neq -2det(B)$det(C) ≠ −2det(B).
- [ ] $det(C) = -2det(A)$det(C) = −2det(A).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = -det(B)$det(A) = −det(B).
$det(C) \neq -2det(B)$det(C) ≠ −2det(B).
$det(C) = -2det(A)$det(C) = −2det(A).

(Based on the below information answer questions 8 and 9).

Let AA be a real $3 \times 3$3 × 3 matrix as given below:

$$A = [C_1 \; C_2 \; C_3]$$ A = [C1C2C3]

where C_i Ci represents the ii-th column of the matrix AA. Now let us consider the following set of matrices.

- $A_1 = [C_1\ C_2 + 5C_3\ C_3]$A1 = [C1C2 + 5C3C3]
- $A_2 = [C_1 + C_2 + C_3\ C_2\ C_3]$A2 = [C1 + C2 + C3C2C3]
- $A_3 = [C_1\ C_2 + 5C_3\ 0]$A3 = [C1C2 + 5C30]
- $A_4 = [C_1 + C_2\ C_2 + C_3\ C_3 + C_1]$A4 = [C1 + C2C2 + C3C3 + C1]

1 point

Which of the matrices have the same determinant as that of AA?

☐ A_1 A1

☐ A_2 A2

☐ A_3 A3

☐ A_4 A4

No, the answer is incorrect.
Score: 0
Accepted Answers:

A_1 A1

A_2 A2

1 point

Choose the set of correct options.

☐ $det(A_3)$det(A3) cannot be determined from the given information.

☐ $det(A_3) = 0$det(A3) = 0

☐ $det(A_4) = 2\ det(A)$det(A4) = 2 det(A)

☐ $det(A_2) = 3\ det(A)$det(A2) = 3 det(A)

No, the answer is incorrect.
Score: 0
Accepted Answers:

$det(A_3) = 0$det(A3) = 0

$det(A_4) = 2\ det(A)$det(A4) = 2 det(A)

If the diagonal entries of a $3 \times 3$3 × 3 lower triangular matrix AA are 1,2, and 3, then find the sum of roots of the equation $det(A - xI) = 0$det(A − xI) = 0, where II is the identity matrix of order 3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 6

1 point

Check Answers

Your score is: 0/10